

Underwater Odyssey: Immersive VR Swimming Adventure

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Figure 1: Screenshot of the VR underwater moving experience in our Demo

Abstract

The Underwater Odyssey provides participants with an immersive journey into the ocean’s depths through virtual reality interaction. In contrast to common VR locomotion methods such as teleportation or joystick, our work explores a novel approach: using our proposed stroking controller method to simulate swimming movements. This interactive method aligns with the ocean theme of our project, offering users with an experience similar to navigating underwater environments. Our Demo creates a unique way of moving that contributes to VR mobile interaction.

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1 Introduction

The quest for an authentic and immersive VR experience has been a driving force in the field of human-computer interaction. While terrestrial VR navigation relies on joystick and teleportation systems, these methods fail to simulate the hydrodynamic dynamics of aquatic environments, often causing motion sickness and breaking immersion. Recent efforts to improve underwater realism include specialized walking-in-place systems with leg sensors [Tan et al.

2022] and full-body aquatic simulations in physical pools [Fauville et al. 2024]. Although innovative, these approaches face inherent limitations, water-based setups introduce practical constraints [Jain et al. 2016], and activity-specific implementations lack generalizability for broader underwater navigation. While recent work has explored pseudo-haptics to simulate localized drag forces on limbs in stationary underwater settings [Kang et al. 2019], a general solution for simulating full-body propulsion and navigation through swimming motions remains an open challenge.

This Demo introduces an innovative approach to underwater VR locomotion that aims to address these limitations. We introduce stroking controllers, enabling users to safely mimic swimming motions without leaving their spot, thus providing a more natural and immersive interaction with virtual underwater environments.

2 Design And User Experience

Underwater VR movement fundamentally differs from terrestrial locomotion by enabling full 3D navigation along the X-Y-Z axes. This demo delivering a natural underwater experience, realistically simulating water resistance to create authentic swimming sensations.

Participants begin their VR journey with an interactive tutorial by pressing the VR controller buttons to set a starting position and then pulling back to execute a swimming motion that teaches them the fundamental operation to mimic swimming. Following the tutorial, users embark on their adventure, starting within a submarine, where they interact with the onboard machinery to locate a virtual tool, a simulated pistol for their journey; upon exiting the submarine, they are greeted by the expansive underwater world. The core mission involves collecting special flames scattered throughout the scene, as these flames are instrumental in activating three strategically placed lighthouses within the environment. After activating two lighthouses, users are led to an underwater

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palace where they face off against an octopus boss; this battle is a test of the swimming techniques they've acquired, as they must evade the boss's attacks and ultimately defeat it. Victory is marked by the activation of the third lighthouse, which triggers a timeline reboot; with the completion of the mission, users are rewarded with a spectacular view of the city before its submersion.



Figure 2: (A) Underwater flame and lighthouses. (B) Octopus in the underwater palace. (C) The glorious city before it sank.

3 Implementation

Our core idea lies in transforming the VR controller into a stroking controller, specifically designed to replicate the act of swimming. This approach encourages users to physically mimic swimming motions, providing a more intuitive and immersive underwater VR experience. Implementing the stroking controller's movement method in this demonstration relies on calculating the positional delta between two points. When the controller is positioned in front of the user, pressing a button on the controller records its initial position. Subsequently, as the user strokes the controller backward without releasing the button, the endpoint of the stroke is noted upon button release, signifying the completion of a stroking action. Swimming velocity is calculated by dividing the distance between the initial and final positions by the duration of a button press.

To further enhance the realism, our system incorporates a simulation of the inertia and resistance forces that are inherent to underwater movement. The drag force is calculated using the principles of fluid dynamics:

$$F_d = -\frac{1}{2}\rho v^2 A C_d \quad (1)$$

where fluid density ρ , user velocity v , cross-sectional area A , and drag coefficient (C_d collectively simulate water resistance. The influence of the resistance on the user's motion is integrated over time to compute the updated velocity:

$$\vec{v}(t + \Delta t) = \vec{v}(t) - \frac{F_d}{m} \Delta t \quad (2)$$

Here virtual mass m and timestep Δt govern the progressive velocity attenuation mirroring real-world swimming dynamics. The introduced kinematic delay between physical gestures and virtual limb motion simulates aquatic inertia, creating congruent visual feedback that preserves immersion while reducing motion sickness.

4 Evaluation And Results

To objectively validate our locomotion design, we developed a dedicated underwater runway testing environment with floating checkpoints, where twenty-two participants (aged 18–35) compared three movement methods: teleportation, swimming, and joystick control. After experiencing each method, participants completed a questionnaire using a 7-point Likert scale to assess realism, immersion, comfort, and motion sickness. Participants received standardized

training through instructional videos to ensure consistent comparison.

Analysis of the user study data, as illustrated in Figure 3, Results demonstrated clear preference for swimming gestures, which outperformed other methods in immersion and realism metrics. While teleportation showed the lowest motion sickness incidence, it received significantly poorer ratings for environmental presence. The Swimming method notably reduced discomfort compared to joystick controls, achieving a balanced profile of moderate physical comfort and superior experiential quality.

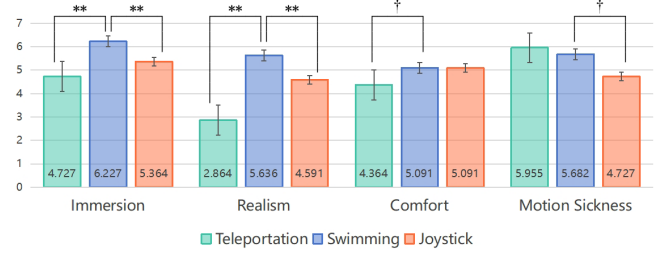


Figure 3: Locomotion experience Average point for the three locomotion methods.(†: $p < .1$, **: $p < .01$)

5 Summary

This paper presents an innovative approach to enhancing the virtual reality experience in underwater settings through the development of stroking controllers that mimic swimming actions. This method significantly increases the authenticity and immersion of underwater navigation in VR, creating a more engaging and enjoyable user experience. The introduction of these new locomotion techniques aims to address and reduce common issues such as motion sickness by synchronizing physical movements with virtual perceptions. We anticipate that this demo will offer valuable insights into embodied interaction design and inspire further exploration of intuitive locomotion techniques for immersive XR environments.

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